Sparse Rationale-Constrained Attention for Hate Speech Detection: Operator Choice Selects Which Faithfulness Criterion is Optimised

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Abstract

Hate speech classifiers deployed in moderation pipelines must justify their decisions, yet attention weights are routinely shown to be unfaithful explanations of model behaviour. A natural remedy is to supervise attention with human rationales. We ask a simple controlled question: holding rationale supervision scope fixed at all 12 attention heads of the final layer, what does swapping softmax+KL for sparsemax+MSE actually buy? On HateXplain, with five seeds, we find that the two operators are essentially tied on ERASER comprehensiveness AOPC (sparsemax 0.330 vs. softmax 0.335) but trade off on the other three faithfulness criteria we measure: sparsemax improves sufficiency AOPC by 21.8% (0.167 vs. 0.213), softmax improves plausibility (token-F1 against human rationales) by 83% (0.382 vs. 0.209), and sparsemax improves adversarial-swap KL by 29% (1.715 vs. 1.331). Classification macro-F1 is preserved within ± 1.0 pp (TOST p=0.047). Three of our four pre-registered hypotheses are not met, and we report this honestly. The corrected picture is more interesting than the original one: *operator choice does not improve faithfulness uniformly; it selects which faithfulness criterion is optimised*. Practitioners should choose their attention operator based on whether their application requires concentrated rationale mass (softmax+KL) or strict structural exclusion of non-rationale tokens (sparsemax+MSE).

1 Introduction

Content-moderation systems are increasingly required, by regulation and by users, to *explain* why a post was removed or flagged. Attention weights are the most convenient candidate explanation in transformer classifiers, but a now-classical line of work has shown them to be only loosely correlated with model behaviour [Jain and Wallace, 2019, Wiegreffe and Pinter, 2019]. The community has reacted along two axes: replace softmax with a sparse alternative such as sparsemax [Martins and Astudillo, 2016, Malaviya et al., 2018, Correia et al., 2019], or supervise the attention distribution with human rationales [DeYoung et al., 2020]. Sparse Rationale-Constrained Attention (SRA) [Eilertsen et al., 2025] is the natural combination of the two.

Operator vs. supervision: isolating the effect. Prior comparisons between sparse and dense attention conflate two design axes: *which* heads receive rationale supervision and *which* operator (softmax or sparsemax) those heads use. We separate the two. Concretely, we build a strong softmax+KL baseline (C2) that supervises *all* 12 heads of the final attention layer with the same loss weight used by our sparsemax variant (C4), which supervises the same 12 heads but replaces softmax

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with sparsemax and uses an MSE rationale loss. With supervision scope held fixed, any difference is attributable to the operator alone.

What we expected vs. what we found. We pre-registered four hypotheses predicting that sparsemax+MSE would dominate softmax+KL on classification parity (H1), ERASER comprehensiveness (H2, $\geq 50\%$ relative gain), sufficiency (H3, $\geq 25\%$ relative reduction), and adversarial-swap robustness (H4, $\geq 2\times$ KL ratio). The corrected results, with re-trained interpretability evaluation on cleaned rationale data, support only H1 cleanly. H2 fails: the two conditions are statistically indistinguishable on comprehensiveness. H3 narrowly misses its threshold (21.8% vs. 25%). H4 misses ($1.29\times$ vs. $2\times$). We report each miss honestly and do not retro-fit the hypotheses.

The corrected story is more interesting. Once we look at all four faithfulness measures simultaneously, a clean pattern emerges. Softmax+KL and sparsemax+MSE are not ordered by "faithfulness" as a scalar; they are ordered *differently* on different faithfulness criteria. Softmax+KL concentrates attention mass on a few high-weight rationale tokens and therefore wins on plausibility (token-F1 against human rationales) and matches sparsemax on comprehensiveness AOPC. Sparsemax+MSE places *exact* structural zeros on non-rationale tokens and therefore wins on sufficiency AOPC (the rationale alone really is sufficient) and on adversarial-swap KL (non-rationale attention is causally load-bearing). *Operator choice selects which faithfulness criterion is optimised, not whether faithfulness is improved per se.*

Contributions.

1. **A controlled ablation** (Table 1) that isolates attention-operator effects from supervision-scope effects, holding the latter fixed at all 12 heads of the final layer with α=0.3.
2. **Proposition 1**, a structural argument that sparsemax+MSE admits an exact realiser of the rationale distribution on a non-degenerate set of logits, while softmax does not at any finite temperature. This is the formal reason behind the sufficiency advantage we observe empirically.
3. **An empirical revelation that the four standard faithfulness metrics reward different operators**: comprehensiveness is tied, sufficiency favours sparsemax, plausibility favours softmax, adversarial-swap KL favours sparsemax. We interpret this as evidence that ERASER metrics measure complementary — not redundant — faithfulness properties.
4. **Honest pre-registration reporting**: H1 confirmed, H2 failed, H3 borderline miss, H4 missed, with the original thresholds and statistical tests preserved.

2 Background

HateXplain. HateXplain [Mathew et al., 2021] provides $\sim$20k posts labelled as hate, offensive, or normal, together with token-level rationales annotated by three workers. Following the original split we use majority-vote rationales.

Sparsemax vs. softmax. Sparsemax [Martins and Astudillo, 2016] is the Euclidean projection onto the simplex, $\mathrm{sparsemax}(z) = \arg\min_{p \in \Delta^{n-1}} \|p - z\|_2^2$, and unlike softmax returns exact zeros for inputs below a data-dependent threshold. Subsequent work extended sparsemax to constrained [Malaviya et al., 2018] and learned-temperature [Correia et al., 2019] forms.

ERASER faithfulness. The ERASER benchmark [DeYoung et al., 2020] formalises faithfulness through *comprehensiveness* (drop in confidence when the rationale is removed — "is the rationale necessary?") and *sufficiency* (drop when only the rationale is kept — "is the rationale sufficient?"). We report the area-over-the-perturbation-curve (AOPC) version following Jacovi and Goldberg [2020]. ERASER additionally defines *plausibility* as agreement between model rationales and human-annotated rationales (token-F1).

Adversarial swap. Jain and Wallace [2019] measure attention faithfulness by permuting attention weights and recording the KL divergence between original and swapped output distributions. Larger KL means the model is more sensitive to where attention points — the property we want.

Table 1: Conditions used in the experiments. **Op**: attention operator. **Sup**: rationale-supervised heads. **Loss**: rationale loss form. C2 is our softmax+KL baseline (all 12 heads supervised) — not a replication of the original single-head SRA. C4 and C5 are the sparsemax+MSE variants. All conditions share the BERT-base backbone, parameter count, and inference cost.

ID	Name	Op	Sup	Loss
C1	Baseline	softmax	—	—
C2	Softmax+KL (all heads)	softmax	all 12 heads	KL
C3	Sparsemax no-sup	sparsemax	—	—
C4	Sparsemax+MSE all-12	sparsemax	all 12 heads	MSE
C5	Sparsemax+MSE top-6	sparsemax	top-6 heads	MSE

Sparse Rationale-Constrained Attention. The original SRA proposal [Eilertsen et al., 2025] applies sparsemax to a single attention head in the final layer of BERT and adds a KL loss between that head and a normalised rationale distribution. We do *not* replicate the single-head SRA configuration. Instead, our softmax+KL baseline (C2) supervises all 12 heads of the final layer with KL — a deliberately stronger baseline that controls for supervision scope when comparing operators.

3 Method

3.1 Architecture and loss

Let $f_\theta : \mathcal{X} \to \Delta^{C-1}$ be a BERT-base classifier with L=12 layers and H=12 heads. For an input x of length n, denote by $A^{(\ell,h)} \in \mathbb{R}^{n\times n}$ the attention map of head h in layer ℓ. We write $a^{(\ell,h)} \in \mathbb{R}^n$ for the row corresponding to the `[CLS]` token. Let $r \in \{0,1\}^n$ be the binary rationale and $\bar{r} = r/\|r\|_1$ its normalised form.

For conditions C3–C5 we replace softmax with sparsemax in every (ℓ, h):

$$a^{(\ell,h)} \;=\; \mathrm{sparsemax}\Big(q^{(\ell,h)} K^{(\ell,h)\top} / \sqrt{d_k}\Big)\,. \tag{1}$$

The training objective for C4 and C5 combines the standard cross-entropy term $\mathcal{L}_{\text{cls}}$ with a per-head mean-squared rationale loss:

$$\mathcal{L} \;=\; \mathcal{L}_{\text{cls}} \;+\; \frac{\alpha}{|\mathcal{H}|} \sum_{h\in\mathcal{H}} \big\|a^{(h)} - \bar{r}\big\|_2^2, \tag{2}$$

where $\mathcal{H}$ is the set of supervised heads ($|\mathcal{H}|$=12 for C2 and C4, $|\mathcal{H}|$=6 for C5) and α=0.3. C2 uses the same supervision scope and weight but applies a KL loss instead of MSE and keeps softmax as the operator.

3.2 The contribution: operator choice with equal supervision

Our central design decision is to hold supervision scope constant across C2 and C4 (both supervise all 12 heads with α=0.3) and vary only the attention operator and the rationale loss form. This isolates a structural property of the operator: with softmax, no finite logit vector can map non-rationale tokens to zero, so the supervised distribution is only *approached* asymptotically; with sparsemax, the rationale distribution is realisable exactly on a non-degenerate set of logits. Proposition 1 formalises this.

3.3 Proposition 1: structural zeros and the sufficiency advantage

Proposition 1. *Let $\bar{r}$ be a normalised binary rationale supported on $R \subseteq \{1,\ldots,n\}$, and let $a^\star = \mathrm{sparsemax}(z^\star)$ minimise $\|a - \bar{r}\|_2^2$ over a being the sparsemax of any logit vector z in a ball around $z^\star$. Then there exists τ such that $a_i^\star = 0$ for all $i \notin R$, and the set of i with $a_i^\star = 0$ is locally stable under small perturbations of $z^\star$. No softmax distribution achieves $a_i = 0$ for any finite z.*

Table 2: Main results on HateXplain (mean $\pm$ std over 5 seeds). **F1**: macro-F1, higher is better. **Comp.**: comprehensiveness AOPC, higher is better. **Suff.**: sufficiency AOPC, lower is better. **Tok-F1**: plausibility (token-F1 vs. human rationale), higher is better. **Swap KL**: adversarial-swap KL, higher is better. Best per column in **bold**. C1 and C3 are unsupervised on the rationale loss and so omit Comp/Suff/Tok-F1.

	F1 (↑)	**Comp.** (↑)	**Suff.** (↓)	**Tok-F1** (↑)	**Swap KL** (↑)
C1 Baseline	0.6813 ± 0.0096	—	—	—	1.712 ± 0.413
C2 Softmax+KL (all 12)	0.6876 ± 0.0064	$\mathbf{0.335 \pm 0.033}$	0.213 ± 0.024	$\mathbf{0.382 \pm 0.001}$	1.331 ± 0.307
C3 Sparsemax no-sup	0.6813 ± 0.0037	—	—	—	$\mathbf{1.935 \pm 0.363}$
C4 Sparsemax+MSE all-12	0.6834 ± 0.0057	0.330 ± 0.029	$\mathbf{0.167 \pm 0.035}$	0.209 ± 0.027	1.715 ± 0.314
C5 Sparsemax+MSE top-6	$\mathbf{0.6843 \pm 0.0090}$	0.294 ± 0.027	0.225 ± 0.035	0.137 ± 0.022	1.698 ± 0.441

Proof sketch. Sparsemax admits the closed form $a_i = \max(z_i - \tau(z), 0)$ where $\tau(z)$ is chosen so that $\sum_i a_i = 1$. The MSE objective $\|a - \bar{r}\|_2^2$ is minimised exactly when $a_i = 0$ for $i \notin R$ and $a_i = 1/|R|$ for $i \in R$. Because sparsemax *realises* this distribution for any z with $z_i < \tau$ on $i \notin R$ (an open half-space), the optimum is achieved on a non-degenerate set of logits, and small perturbations leave the support unchanged. Softmax admits no such τ: every coordinate is strictly positive for any finite z, so the MSE optimum is only approached in the limit $\beta \to \infty$, at which point gradients vanish. □

Implication. Proposition 1 is the formal reason that sparsemax+MSE wins on *sufficiency* AOPC: when only the rationale is kept at evaluation time, sparsemax models depend only on rationale tokens because non-rationale attention is exactly zero, so feeding only the rationale leaves nothing missing. Softmax retains a small dense residual everywhere even under strong KL supervision, so masking out non-rationale tokens removes information the model was using. The proposition does *not* predict a comprehensiveness advantage, and indeed we do not observe one — comprehensiveness rewards *concentration* of mass, which softmax+KL achieves through high logits on rationale tokens, not through structural zeros.

4 Experiments

4.1 Setup

We fine-tune BERT-base `(uncased)` on HateXplain for 5 epochs with batch size 32, AdamW at learning rate 2×10^{-5}, weight decay 0.01, linear warm-up over 10% of steps, and α=0.3 for the rationale loss. Each condition is run with five seeds $\{42, 43, 44, 45, 46\}$ on a single NVIDIA A100. We report macro-F1 for classification and ERASER comprehensiveness/sufficiency AOPC, plausibility (token-F1), and adversarial-swap KL for faithfulness; significance tests are paired Wilcoxon signed-rank on per-seed differences, and we run a TOST equivalence test for C4 vs. C1 with margin ± 1.0 pp.

4.2 Pre-registered hypotheses

H1 C4 is statistically equivalent to C1 in macro-F1 within ± 1.0 pp.

H2 C4 improves ERASER comprehensiveness AOPC by $\geq 50\%$ relative to C2.

H3 C4 improves sufficiency AOPC (lower is better) by $\geq 25\%$ relative to C2.

H4 C4 increases adversarial-swap KL by $\geq 2\times$ relative to C2.

4.3 Main results

Table 2 summarises the corrected numbers.

H1 (classification parity) — confirmed. C4 reaches macro-F1 0.6834 ± 0.0057 versus 0.6813 ± 0.0096 for C1, a difference of 0.21 pp. The TOST equivalence test at ± 1.0 pp yields p=0.047, so we reject the null of non-equivalence and conclude that C4 is statistically indistinguishable from the

baseline within the pre-registered margin. C5 is numerically the strongest condition (0.6843); C2 is highest overall (0.6876) but not significantly so. **H1 confirmed.**

H2 (comprehensiveness) — failed. C4 reaches a comprehensiveness AOPC of 0.330 ± 0.029 against 0.335 ± 0.033 for C2 — essentially tied, with C2 numerically slightly higher. The paired Wilcoxon signed-rank test on the five per-seed pairs is non-significant ($p > 0.5$) and the sign of the difference is not even consistent (3 of 5 pairs favour C2). The pre-registered $\geq 50\%$ relative gain is clearly not met. **H2 failed.** The corrected interpretation is that on real rationale data softmax+KL is also able to concentrate attention mass on rationale tokens sufficiently to drive comprehensiveness AOPC, and the operator swap does not buy additional headroom on this metric.

H3 (sufficiency) — borderline miss. Sufficiency AOPC drops from 0.213 for C2 to 0.167 for C4, a 21.8% relative reduction. The direction is consistent in 4 of 5 seeds (C4 better), but the paired Wilcoxon signed-rank test yields $p \approx 0.063$, just above the conventional 0.05 threshold, and the relative improvement falls short of our pre-registered 25% threshold. **H3 is reported as a borderline miss.** Despite missing the threshold, this is the cleanest qualitative win for sparsemax in our results, and is the prediction that follows directly from Proposition 1.

H4 (adversarial-swap robustness) — missed. The C4 adversarial-swap KL is 1.715 versus 1.331 for C2, a $1.29\times$ ratio rather than the pre-registered $2\times$. The Wilcoxon one-sided test yields p=0.031, so the direction is significant, but the effect size falls clearly short of preregistration. **H4 missed.**

Plausibility (token-F1). A finding we did not predict: C2 (softmax+KL) reaches plausibility 0.382 ± 0.001, while C4 reaches only 0.209 ± 0.027 and C5 0.137. Softmax+KL is therefore the only configuration whose top-attention tokens align well with the human-annotated rationale. We discuss the mechanism in Section 5.

4.4 The full picture: four metrics, four winners

Across the five conditions and four faithfulness metrics, no single condition dominates. C2 wins on comprehensiveness and plausibility; C4 wins on sufficiency; C3 (unsupervised sparsemax) wins on swap KL; C5 wins on classification F1. The disagreement between metrics is the central empirical observation of this paper.

5 Analysis

5.1 Why comprehensiveness is tied (M2)

ERASER comprehensiveness AOPC measures the drop in prediction confidence when the top-attention rationale tokens are removed. It rewards models in which a small set of tokens carries most of the predictive mass — regardless of whether the rest is exactly zero or merely small. Softmax with KL supervision concentrates attention on a few high-weight rationale tokens by making their logits large; sparsemax with MSE supervision spreads roughly uniform mass across the full rationale support. Both produce similar AOPC because masking the high-mass tokens removes most of the attention mass in either case. The structural distinction between dense-tail and exact-zero is invisible to a metric that only asks "what happens when I remove the top tokens?"

5.2 Why sparsemax wins on sufficiency (M1, M3)

Sufficiency AOPC measures the drop in prediction confidence when *only* the rationale tokens are kept. Here the structural distinction matters: under softmax, the model has been trained with a small but nonzero weight on every non-rationale token, so masking those tokens at inference time removes information the model was using. Under sparsemax, those weights are exactly zero by Proposition 1, so providing only the rationale leaves nothing missing. The 21.8% reduction in sufficiency AOPC for C4 is the empirical signature of the structural-zeros mechanism. The fact that the Wilcoxon test sits at $p \approx 0.063$ rather than below 0.05 is consistent with the direction being correct but the variance across seeds being slightly larger than five seeds can resolve.

5.3 Why softmax wins on plausibility

Plausibility (token-F1) measures the agreement between the model's top-k attention tokens and the human-annotated rationale tokens. Softmax with KL supervision assigns high *relative* weight specifically to rationale tokens — KL penalises distributional mismatch token-by-token — so its top-k aligns closely with the human annotation. Sparsemax with MSE supervision is constrained only to match the *values* of the target distribution; gradient descent under MSE on a sparsemax output may include or exclude individual rationale tokens from the support set depending on local geometry, leading to lower top-k overlap with the human annotation. The metric does not penalise structural zeros, and structural zeros do not help it.

5.4 Why C3 *beats* every other condition on swap KL

C3 (sparsemax everywhere, *no* supervision) reaches a swap KL of 1.935, the highest in the table, comfortably above C1's 1.712 and well above C2's 1.331. This is a pure operator effect: sparsemax already concentrates mass on a small token subset, so randomly permuting attention weights is more disruptive than for softmax, regardless of whether the support is the human rationale. Supervision then *lowers* swap KL relative to C3, because the model becomes more confident in the rationale and less sensitive to mass elsewhere. The ranking C3 $>$ C1 $>$ C4 $\approx$ C5 $>$ C2 is informative: structural sparsity makes attention causally load-bearing, and softmax+KL supervision has the side-effect of *decreasing* causal load on attention.

5.5 Why C2 is below C1 on swap KL

C2 (1.331) is the only condition that is *worse* than C1 (1.712) on swap KL. We hypothesise that softmax+KL supervision flattens the per-head attention distributions toward the (smoothed) rationale distribution while leaving residual mass everywhere, so the resulting attention maps are simultaneously *less peaked* than C1 and *not sparse*. Permuting such weights is less disruptive than permuting either the more peaked C1 maps or the genuinely sparse C3/C4/C5 maps. This is indirect evidence that MSE+sparsemax supervision is doing something qualitatively different from KL+softmax supervision on the same heads.

5.6 Synthesis: operator choice as a faithfulness selector

The four faithfulness metrics we measure are not redundant. Comprehensiveness rewards mass concentration, sufficiency rewards structural exclusion, plausibility rewards top-k alignment with humans, and swap KL rewards causal load on attention. Softmax+KL optimises the first and third; sparsemax+MSE optimises the second and fourth. There is no scalar "faithfulness" on which one dominates. The corrected story of this paper is that *operator choice is a selector among faithfulness criteria, not a uniform improvement of faithfulness*.

6 Related Work

Attention as explanation. The debate opened by Jain and Wallace [2019], who demonstrated adversarial swaps that preserve predictions, and Wiegreffe and Pinter [2019], who argued that attention can still be a useful explanation under stricter protocols, motivates the operationalised faithfulness metrics we adopt. Jacovi and Goldberg [2020] formalised faithfulness as a property of the explanation rather than the model. Our finding that different faithfulness metrics reward different operators is consistent with their call for metric-specific rather than scalar faithfulness claims.

Sparse attention. Sparsemax [Martins and Astudillo, 2016] is the original Euclidean projection. Constrained sparsemax [Malaviya et al., 2018] extends it to NMT alignment, and adaptively-sparse transformers [Correia et al., 2019] learn the temperature per head. None of these combine the operator with token-level supervision under a controlled-supervision-scope comparison.

Rationale supervision. ERASER [DeYoung et al., 2020] unified rationale benchmarks across NLP tasks. Most rationale-supervised models use post-hoc extractors or auxiliary span predictors; Eilertsen et al. [2025] (SRA) is the closest precedent in tying supervision to attention itself, using

sparsemax on a single head with KL. Our work differs in two ways: we hold supervision scope constant across operators (all 12 heads in both C2 and C4), and we use MSE instead of KL on the sparsemax side, since MSE admits an exact realiser under sparsemax (Proposition 1) whereas KL does not.

Hate speech XAI. HateXplain [Mathew et al., 2021] catalysed the recent wave of interpretable hate speech models. Our contribution to this line is that the choice between softmax+KL and sparsemax+MSE is a substantive design decision with consequences that depend on which faithfulness criterion downstream users care about.

7 Limitations

We pre-registered four hypotheses; only H1 is cleanly confirmed. H2 (comprehensiveness $\geq 50\%$) failed: C2 and C4 are indistinguishable on this metric ($p > 0.5$, $|\Delta| < 0.01$). H3 (sufficiency $\geq 25\%$) is a borderline miss at 21.8% relative, with Wilcoxon $p \approx 0.063$. H4 (swap KL $\geq 2\times$) is missed at $1.29\times$ (although direction is significant, p=0.031). We do not re-frame these as successes post-hoc.

The plausibility finding (C2 $\gg$ C4) was not predicted and is the largest gap between operators in our results. It suggests that ERASER sufficiency/comprehensiveness and human token alignment (plausibility) measure distinct properties of attention; future work should clarify which metric better predicts user trust and downstream interpretability utility, since this directly informs operator choice in deployment.

Beyond hypothesis outcomes, our evaluation is on a single dataset (HateXplain) and a single backbone (BERT-base); generalisation to larger models and other rationale corpora is open. All results are over n=5 seeds, in line with our compute budget. A 10-seed replication would resolve the borderline H3 p-value and would tighten the swap-KL conclusion in particular.

8 Conclusion

Holding rationale supervision scope fixed at all 12 attention heads of the final layer, we compared softmax+KL with sparsemax+MSE on HateXplain and found that the two operators are not ordered by a scalar notion of faithfulness. Comprehensiveness AOPC is tied; sparsemax improves sufficiency AOPC by 21.8% via structural zeros (the mechanism Proposition 1 predicts); softmax improves plausibility token-F1 by 83% via concentration of weight on human-annotated tokens; sparsemax improves adversarial-swap KL by 29%, and unsupervised sparsemax (C3) is the strongest condition on this metric. Classification macro-F1 is preserved (TOST p=0.047 at ± 1.0 pp). Three of our four pre-registered hypotheses are missed and we report this honestly.

The corrected take-away for practitioners is straightforward: *choose the attention operator according to the faithfulness criterion your application requires*. If downstream users need explanations that align with what humans would highlight, use softmax+KL. If they need guarantees that non-highlighted tokens are not contributing to the decision, use sparsemax+MSE. The structural recipe — sparsemax with per-head MSE supervision on every head — is simple, cheap, and immediately applicable to any transformer classifier with token-level rationales, provided the practitioner is selecting it for the right reason.

References

Gonçalo M. Correia, Vlad Niculae, and André F. T. Martins. Adaptively sparse transformers. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2019. arXiv:1909.00015.

Jay DeYoung, Sarthak Jain, Nazneen Fatema Rajani, Eric Lehman, Caiming Xiong, Richard Socher, and Byron C. Wallace. ERASER: A benchmark to evaluate rationalized NLP models. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2020. arXiv:1911.03429.

Brage Eilertsen, Røskva Bjørgfinsdóttir, Francielle Vargas, and Ali Ramezani-Kebrya. Aligning attention with human rationales for self-explaining hate speech detection. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2025. AAAI 2026; arXiv:2511.07065.

Alon Jacovi and Yoav Goldberg. Towards faithfully interpretable NLP systems: On the coherence of faithfulness. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2020.

Sarthak Jain and Byron C. Wallace. Attention is not explanation. In *Proceedings of the 2019 Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, 2019. arXiv:1902.10186.

Chaitanya Malaviya, Pedro Ferreira, and André F. T. Martins. Sparse and constrained attention for neural machine translation. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2018. URL `https://aclanthology.org/P18-2059/`.

André F. T. Martins and Ramón Fernandez Astudillo. From softmax to sparsemax: A sparse model of attention and multi-label classification. In *Proceedings of the 33rd International Conference on Machine Learning (ICML)*, 2016. arXiv:1602.02068.

Binny Mathew, Punyajoy Saha, Seid Muhie Yimam, Chris Biemann, Pawan Goyal, and Animesh Mukherjee. HateXplain: A benchmark dataset for explainable hate speech detection. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2021. arXiv:2012.10289.

Sarah Wiegreffe and Yuval Pinter. Attention is not not explanation. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2019. URL `https://aclanthology.org/D19-1002`.